Part 5

Suppose we have a microservice architecture with many different services. We also have an app that gets data from these different services. The app runs on both Android and IOS. The problem we face is that when we start this app, the first page on the app needs to show data that comes from 15 different microservices. The time to retrieve all this data from all 15 microservices takes 21 seconds, which is too slow and not acceptable.

The app should get all the required data as fast as possible. It should not take more than 2.5 seconds to retrieve all necessary data.

- Caching does not solve our problem because the data cannot be stale longer than 1 hour.
- Calling the services asynchronous is also not solving our problem because the calls we make to some services depend on the data we get back from other services

Explain how we can solve this problem?

Answer:

Use **CQRS (Command Query Responsibility Segregation)** with a dedicated **read/query service** for the mobile home page.

Instead of the Android/iOS app calling 15 microservices, it makes **one request** through the API Gateway to the query service. Each microservice publishes events when its data changes, and the query service updates a denormalized read database containing all data needed for the first screen.

This makes the first-page request fast because it reads pre-combined data rather than waiting for 15 dependent service calls. The read model is eventually consistent, but this is acceptable because data may be up to one hour old. CQRS is suitable here because read performance is critical.